Week 8

Introduction

In the rapidly evolving landscape of power electronics, particularly for demanding applications like electric vehicle charging infrastructure, the efficiency and reliability of DC-DC converters are paramount. As the drive for faster charging and more compact systems intensifies, engineers face the challenge of increasing switching frequencies without incurring prohibitive energy losses. This week's lessons delve into advanced Pulse Width Modulation (PWM) strategies and the sophisticated Phase-Shifted Full Bridge (PSFB) converter topology, which are specifically designed to mitigate these losses through "soft switching" techniques.

Understanding these advanced modulation strategies is crucial for optimizing converter performance. By carefully controlling the turn-on and turn-off sequences of power semiconductor devices, it's possible to achieve Zero Voltage Switching (ZVS) or Zero Current Switching (ZCS), significantly reducing energy dissipation during transitions. We explore the operational nuances of different PWM types and the PSFB converter, highlighting how they leverage parasitic elements like leakage inductance and device capacitances to enable soft switching. Furthermore, we introduce the critical technique of small-signal modeling, an indispensable tool for analyzing the dynamic behavior of these converters and designing robust control systems that can maintain stable output voltage despite varying input conditions and loads.

Core Concepts

Term	Definition	Significance
`Zero Voltage Switching (ZVS)`	A soft-switching technique where a power semiconductor switch turns on or off when the voltage across it is zero.	Significantly reduces switching losses, improves efficiency, and allows for higher switching frequencies, leading to smaller component sizes.
`Leading Leg`	One half-bridge of the full-bridge converter, whose switching transitions often initiate the power transfer cycle.	Its ZVS operation is generally robust across varying load conditions, contributing significantly to overall converter efficiency.

Term	Definition	Significance
`Lagging Leg`	The other half-bridge of the full-bridge converter, whose switching transitions often follow the leading leg.	ZVS for this leg is often more challenging to achieve consistently, especially under varying load conditions, impacting overall efficiency.
`Dead Time`	A short, intentional delay between the turn-off of one switch and the turn-on of its complementary switch in a half-bridge.	Essential for allowing parasitic capacitances to charge/discharge, enabling ZVS, and preventing shoot-through currents.
`L_leakage`	The leakage inductance inherent in the primary winding of the high-frequency transformer.	Plays a crucial role in the resonant transitions that enable soft switching, particularly in conjunction with device capacitances.
`Duty Loss (D_loss)`	The reduction in the effective duty cycle of the converter due to the time required for soft-switching transitions.	Directly impacts the output voltage regulation, as the actual output voltage is proportional to (D - D_loss), necessitating compensation.
`Effective Duty Ratio (D_effective)`	The actual duty cycle that contributes to power transfer, calculated as the applied duty ratio minus the duty loss.	This is the true control variable that determines the output voltage, making its accurate modeling essential for precise control.
`Small Signal Model`	A linearized representation of a non-linear system around an operating point, used to analyze its dynamic response to small perturbations.	Indispensable for designing stable and responsive feedback control loops for power converters, allowing for prediction of transient behavior.
`G_VD(s)`		

Term	Definition	Significance
	The transfer function relating the small signal change in output voltage (V_out(s)) to the small signal change in duty ratio (D(s)).	This is the primary goal of small signal modeling, essential for designing feedback controllers to regulate the output voltage.

Detailed Analysis

1. Modulation Strategies for PWM Full Bridge Converter

The week began by exploring fundamental **Modulation Strategies for PWM Full Bridge Converter-II**, focusing on how different PWM techniques impact soft switching.
- **Type-1 PWM**: This strategy involves the simultaneous turn-off of diagonal switches (e.g., S1 and S4).
- It ensures soft turn-off (limited dv/dt) for all switches due to the charging/ discharging of device output capacitances.
- However, it typically results in "hard turn-on" for the complementary switches (e.g., S2 and S3) because resonant oscillations make it difficult to guarantee zero voltage at the turn-on instant. This leads to higher switching losses.
- **Type-2 PWM**: This strategy employs a staggered turn-off of diagonal switches, creating a "zero state" where the primary voltage (VAB) is momentarily zero.
- **Leading Leg (e.g., S1, S2)**: Achieves Zero Voltage Switching (ZVS) turn-on reliably. The primary current (IP), reflecting the output inductor current (IL), provides sufficient energy to discharge the switch capacitances to zero before turn-on.
- **Lagging Leg (e.g., S3, S4)**: ZVS turn-on is more challenging, especially at light loads. During its transition, only the energy stored in the transformer's leakage inductance (L_leakage) is available to discharge the switch capacitances. If the load current is low or L_leakage is small, this energy may be insufficient, leading to hard turn-on.
- Both Type-1 and Type-2 PWM generally ensure soft turn-off for all switches. Type-2 is generally preferred for its improved ZVS capabilities, particularly for the leading leg.

2. Phase Shifted Full Bridge (PSFB) Converter Operation

Building on PWM strategies, the **Phase Shifted Full Bridge (PSFB) Converter- I** and **Phase Shifted Full Bridge (PSFB) Converter-II** lessons introduced a more advanced topology.

- **Evolution to PSFB**: Unlike traditional PWM which varies switch duty cycles, PSFB uses phase-shift modulation. All switches operate with a fixed turn-on time (typically 50% duty cycle), and output voltage regulation is achieved by varying the phase difference between the leading and lagging legs. This method inherently aligns with Type-2 PWM characteristics, making it suitable for ZVS.
- **Operating Modes and ZVS**: The PSFB converter operates through a sequence of modes that facilitate ZVS:
- **Leading Leg ZVS**: When a leading leg switch (e.g., S1) turns off, the primary current (IP) charges/discharges its output capacitances (C1, C2). The large output inductor, reflected to the primary, provides ample energy to ensure the complementary switch's (S2) body diode conducts, allowing S2 to turn on with ZVS. A critical "dead time" is required between S1 turn-off and S2 turn-on, which must be greater than the capacitor discharge time (approx. 2CVin / (NI0)).
- **Lagging Leg ZVS**: When a lagging leg switch (e.g., S4) turns off, the transformer's leakage inductance (L_leakage) resonates with the switch output capacitances (C3, C4). The secondary side diodes become forward biased, effectively short-circuiting the secondary and causing the primary voltage (VAB) to be imposed across L_leakage. The energy stored in L_leakage must be sufficient to discharge C3 to zero for S3 to achieve ZVS turn-on. This is more difficult at light loads where IP is small.
- **Duty Loss (D_loss)**: A significant consequence of ZVS in PSFB is "duty loss." During the resonant transitions for ZVS, the secondary winding is effectively short-circuited (V_R = 0), meaning no power is transferred to the output.
- This period reduces the effective duty cycle, impacting output voltage regulation.
- D_loss is approximated as `4 * L_leakage * n * I0 * F_SW / Vin`, where `n` is the turns ratio, `I0` is output current, `F_SW` is switching frequency, and `Vin` is input voltage.
- D_loss increases with higher L_leakage, output current, and switching frequency, but decreases with higher input voltage. This creates a design trade-off between robust ZVS and minimizing duty loss.

3. Small Signal Modelling of PSFB

The final lessons, **Small Signal Modelling of PSFB-I** and **Small Signal Modelling of PSFB-II**, introduced the crucial technique for analyzing and controlling PSFB converters.
- **Need for Small Signal Modeling**: The PSFB's output voltage is determined by `V0 = N * Vin * D_effective`, where `D_effective = D - D_loss`. Since `D_loss` depends on `IL` (output inductor current) and `Vin`, the relationship between the applied duty ratio (`D`) and `V0` is non-linear and complex. Small-signal modeling linearizes this behavior around an operating point, enabling the design of stable feedback control systems.

- **Buck-Derived Topology**: The PSFB's output stage (inductor and capacitor) behaves like a buck converter. This allows for leveraging established buck converter small-signal models as a starting point, with the input voltage effectively `N*Vin`.
- **Small Signal Approximation**: The methodology involves:
1. Developing an average large-signal model.
2. Introducing small perturbations to steady-state variables (e.g., `D = D + d_tilda`).
3. Linearizing by neglecting products of two small-signal quantities (e.g., `d_tilda * i_l_tilda ≈ 0`).
- **DI and DV Terms**: The change in `D_loss` due to perturbations in `IL` and `Vin` is critical:
- **`DI` term**: A change in `IL` (`i_l_tilda`) causes a change in `D_loss`, which in turn affects `D_effective`. This is captured by `di = - 4 * N * L_leakage * F_SW * i_l_tilda / Vin`. The negative sign indicates that increasing `IL` increases `D_loss`, thus reducing `D_effective`.
- **`DV` term**: A change in `Vin` (`vin_tilda`) also affects `D_loss`. This is captured by `dv = (N * (2 * I0 - V0/L * (1-D) * Ts/2) * L_leakage * vin_tilda / Vin^2) / (Ts/2)`. Increasing `Vin` reduces `D_loss`, thus increasing `D_effective`.
- **Effective Duty Ratio Dynamics**: The total small-signal change in effective duty ratio is `d_effective_tilda = d_tilda + di + dv`. This modified `d_effective_tilda` is then used in the buck converter's small-signal equations, leading to an equivalent circuit with controlled voltage and current sources that account for the `D_loss` effects.
- **Objective: G_VD(s)**: The ultimate goal is to derive the transfer function `G_VD(s) = V_out(s) / D(s)`, which is essential for designing closed-loop controllers to regulate the output voltage.

Key Takeaways

- [ ] Type-1 PWM provides soft turn-off but typically results in hard turn-on for complementary switches.
- [ ] Type-2 PWM improves ZVS, reliably achieving ZVS turn-on for the leading leg due to reflected output inductor energy.
- [ ] ZVS turn-on for the lagging leg in Type-2 PWM is challenging, especially at light loads, as it relies solely on leakage inductance energy.
- [ ] Phase-Shifted Full Bridge (PSFB) converters regulate output voltage by varying the phase shift between switching legs, not individual switch duty cycles.
- [ ] PSFB converters inherently operate like Type-2 PWM, facilitating ZVS for the leading leg, but lagging leg ZVS remains load-dependent.

- [ ] A critical "dead time" is required in PSFB to allow parasitic capacitances to charge/discharge for ZVS.
- [ ] ZVS operation in PSFB leads to "duty loss" (D_loss), which reduces the effective duty cycle and impacts output voltage.
- [ ] Duty loss increases with higher leakage inductance, output current, and switching frequency, but decreases with higher input voltage.
- [ ] Small-signal modeling is essential for PSFB converters to account for the non-linear effects of D_loss and design stable control loops.
- [ ] The effective duty ratio in PSFB is influenced by changes in inductor current (DI term) and input voltage (DV term), which must be incorporated into the small-signal model.

Conclusion

The journey through advanced PWM strategies and the intricacies of the Phase-Shifted Full Bridge converter underscores the sophisticated engineering required for modern power conversion. From understanding the fundamental differences between Type-1 and Type-2 PWM to dissecting the multi-modal operation of the PSFB, we've seen how soft-switching techniques are meticulously employed to enhance efficiency and reduce thermal stress—critical factors in applications like electric vehicle charging. The concept of duty loss, an inherent trade-off for achieving ZVS, highlights the complexities that move beyond idealized converter models.

The introduction to small-signal modeling provides the analytical backbone for tackling these complexities. By linearizing the converter's behavior around an operating point and accounting for the dynamic interplay of duty loss with load current and input voltage, engineers gain the tools to predict system responses and design robust feedback controllers. This foundational knowledge is not merely theoretical; it is directly applicable to developing high-performance, reliable power converters that can meet the stringent demands of future power systems, ensuring stable and precise energy delivery in an increasingly electrified world.